Overview of Naive Bayes classifier

Subject: Naive Bayes classifier

$$P(C_k \mid x) \propto P(C_k) \prod_{i=1}^{n} P(x_i \mid C_k)$$

1.

Constructing a classifier from the probability model The discussion so far has derived the independent feature model, that is, the naive Bayes probability model. The naive Bayes classifier combines this model with a decision rule. One common rule is to pick the hypothesis that is most probable so as to minimize the probability of misclassification; this is known as the maximum a posteriori or MAP decision rule. The corresponding classifier, a Bayes classifier, is the function that assigns a class label y ^ = C k {\displaystyle {\hat {y}}=C_{k}} for some k as follows:

2.

p (x = v ■ C k) = 1 2 π σ k 2 e − (v − μ k) 2 2 σ k 2 {\displaystyle p(x=v\mid C_{k})={\frac {1}{\sqrt {2\pi \sigma _{k}^{2}}}}\,e^{-{\frac {(v-\mu _{k})^{2}}{2\sigma _{k}^{2}}}}}

3.

In practice, there is interest only in the numerator of that fraction, because the denominator does not depend on C {\displaystyle C} and the values of the features x i {\displaystyle x_{i}} are given, so that the denominator is effectively constant. The numerator is equivalent to the joint probability model

4.

evidence = P (male) p (height ■ male) p (weight ■ male) p (foot size ■ male) + P (female) p (height ■ female) p (weight ■ female) p (foot size ■ female) {\displaystyle {\begin{aligned}{\text{evidence}}=P({\text{male}})\,p({\text{height}}\mid {\text{male}})\,p({\text{weight}}\mid {\text{male}})\,p({\text{foot size}}\mid {\text{male}})\\+P({\text{female}})\,p({\text{height}}\mid {\text{female}})\,p({\text{weight}}\mid {\text{female}})\,p({\text{foot size}}\mid {\text{female}})\end{aligned}}}

5.

p (C k ■ x 1 , … , x n) ∝ p (C k , x 1 , … , x n) = p (C k) p (x 1 ■ C k) p (x 2 ■ C k) p (x 3 ■ C k) ■ = p (C k) ∏ i = 1 n p (x i ■ C k) , {\displaystyle {\begin{aligned}p(C_{k}\mid x_{1},\ldots ,x_{n})\varpropto \ &p(C_{k},x_{1},\ldots ,x_{n})\\&=p(C_{k})\ p(x_{1}\mid C_{k})\ p(x_{2}\mid C_{k})\ p(x_{3}\mid C_{k})\ \cdots \\&=p(C_{k})\prod _{i=1}^{n}p(x_{i}\mid C_{k})\,,\end{aligned}}}

6.

Disadvantages Depending on the implementation, Bayesian spam filtering may be susceptible to Bayesian poisoning, a technique used by spammers in an attempt to degrade the effectiveness of spam filters that rely on Bayesian filtering. A spammer practicing Bayesian poisoning will send out emails with large amounts of legitimate text (gathered from legitimate news or literary sources). Spammer tactics include insertion of random innocuous words that are not normally associated with spam, thereby decreasing the email's spam score, making it more likely to slip past a Bayesian spam filter. However, with (for example) Paul Graham's scheme only the most significant probabilities are used, so that padding the text out with non-spam-related words does not affect the detection probability significantly. Words that normally appear in large quantities in spam may also be transformed by spammers. For example, «Viagra» would be replaced with «Viaagra» or «V!agra» in the spam message. The recipient of the message can still read the changed words, but each of these words is met more rarely by the Bayesian filter, which hinders its learning process. As a general rule, this spamming technique does not work very well, because the derived words end up recognized by the filter just like the normal ones. Another technique used to try to defeat Bayesian spam filters is to replace text with pictures, either directly included or linked. The whole text of the message, or some part of it, is replaced with a picture where the same text is "drawn". The spam filter is usually unable to analyze this picture, which would contain the sensitive words like «Viagra». However, since many mail clients disable the display of linked pictures for security reasons, the spammer sending links to distant pictures might reach fewer targets. Also, a picture's size in bytes is bigger than the equivalent text's size, so the spammer needs more bandwidth to send messages directly including pictures. Some filters are more inclined to decide that a message is spam if it has mostly graphical contents. A solution used by Google in its Gmail email system is to perform an OCR (Optical Character Recognition) on every mid to large size image, analyzing the text inside.

7.

p (C k , x 1 , … , x n) = p (x 1 , … , x n , C k) = p (x 1 ■ x 2 , … , x n , C k) p (x 2 , … , x n , C k) = p (x 1 ■ x 2 , … , x n , C k) p (x 2 ■ x 3 , … , x n , C k) p (x 3 , … , x n , C k) = ■ = p (x 1 ■ x 2 , … , x n , C k) p (x 2 ■ x 3 , … , x n , C k) ■ p (x n − 1 ■ x n , C k) p (x n ■ C k) p (C k) {\displaystyle {\begin{aligned}p(C_{k},x_{1},\ldots ,x_{n})&=p(x_{1},\ldots ,x_{n},C_{k})\\&=p(x_{1}\mid x_{2},\ldots ,x_{n},C_{k})\ p(x_{2},\ldots ,x_{n},C_{k})\\&=p(x_{1}\mid x_{2},\ldots ,x_{n},C_{k})\ p(x_{2}\mid x_{3},\ldots ,x_{n},C_{k})\ p(x_{3},\ldots ,x_{n},C_{k})\\&=\cdots \\&=p(x_{1}\mid x_{2},\ldots ,x_{n},C_{k})\ p(x_{2}\mid x_{3},\ldots ,x_{n},C_{k})\cdots p(x_{n-1}\mid x_{n},C_{k})\ p(x_{n}\mid C_{k})\ p(C_{k})\\\end{aligned}}}

8.

p (S ■ D) p (¬ S ■ D) = p (S) p (¬ S) ∏ i p (w i ■ S) p (w i ■ ¬ S) {\displaystyle {p(S\mid D) \over p(\neg S\mid D)}={p(S) \over p(\neg S)}\,\prod _{i}{p(w_{i}\mid S) \over p(w_{i}\mid \neg S)}}

9.

Probabilistic model Abstractly, naive Bayes is a conditional probability model: it assigns probabilities p (C k ■ x 1 , … , x n) {\displaystyle p(C_{k}\mid x_{1},\ldots ,x_{n})} for each of the K possible outcomes or classes C k {\displaystyle C_{k}} given a problem instance to be classified, represented by a vector x = (x 1 , … , x n) {\displaystyle \mathbf {x} =(x_{1},\ldots ,x_{n})} encoding some n features (independent variables). The problem with the above formulation is that if the number of features n is large or if a feature can take on a large number of values, then basing such a model on probability tables is infeasible. The model must therefore be reformulated to make it more tractable. Using Bayes' theorem, the conditional probability can be decomposed as: